A Modular Computational Framework for Section 2 Voting Rights Act Analysis

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Abstract

Section 2 of the Voting Rights Act prohibits redistricting practices that dilute the votes of a racial or ethnic minority. The evidentiary standard set by the Gingles framework requires statistical demonstration of racially polarized voting and of differential electoral opportunity across alternative plans. The components of this evidentiary chain (geospatial data preparation, ecological inference, and ensemble-based outlier analysis) have been developed extensively but largely independently of one another. A modular computational framework that integrates the three components into a single pipeline in which posterior uncertainty from the ecological-inference stage propagates through to plan-level opportunity scoring. We formalize a probabilistic notion of minority electoral opportunity, anchored in a functional that admits an axiomatic justification, and we

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use the Besag-Clifford parallel construction to obtain exact p-values from a Recombination Markov chain regardless of mixing time. A case study of the enacted Texas congressional plan PLANC2333 illustrates how the framework detects systematic dilution of Black and Latino voting power, with Besag-Clifford p-values at or near the resolution floor of 10^{-3} on every opportunity functional examined.

1 Introduction

1.1 Redistricting

Redistricting refers to the periodic redrawing of geographic boundaries that define electoral districts. Because most legislative bodies in democratic systems are comprised of elected representatives from specified constituencies, the lines that split up the districts determine which voters are grouped together, which communities share electoral representation, and which candidates and parties prevail in elections [Cain, 1985].

1.2 The Redistricting Process in the United States

Article I, Section 2 of the U.S. Constitution specifies how seats are apportioned in the House of Representatives among the states [United States, 1787]. After the decennial U.S. census and apportionment is conducted, states usually initiate their redistricting processes, during which they formulate district boundaries accounting for population changes that occured during the past decade [Eckman and Whitaker, 2025].

Although redistricting processes are mostly determined by state laws, congressional district maps are required to comply with the the U.S. Constitution and federal law [Eckman and Whitaker, 2025].

Since the 1964 Wesberry v. Sanders decision, the U.S. Supreme Court has interpreted the Constitution as mandating that every congressional district possess approximately the same number of people [Eckman and Whitaker, 2025]. The population totals are obtained from the U.S. Census; nowadays,all congressional districts in a state have a population difference of zero or at most one [Duchin and Walch, 2022].

Furthermore, the majority of states require that congressional districts be compact and contiguous - that is, that their districts be reasonably shaped and have all their portions be geographically connected, respectively [Duchin and Walch, 2022]. Although not all states explicitly require compactness and contiguity in their Constitutions, this is considered "standard practice" [DeFord et al., 2021]. The U.S. Supreme Court treats compactness and contiguity as traditional redistricting principles and diagnostic signals in racial gerrymandering analysis. For instance, in cases such as Shaw v. Reno [Supreme Court of the United States, 1993], Miller v. Johnson [Supreme Court of the United States, 1995], and Bush v. Vera [Supreme Court of the United States, 1996], lack of compactness and contiguity were cited as evidence of race predominantly determining district lines, which is required to prove racial gerrymandering in violation of the 14th Amendment.

1.3 Gerrymandering

Gerrymandering is a tactic that has been interwoven with United States politics since at least 1812, which is when the term was first coined [Sauer, 1992]. In simplest terms, gerrymandering can be defined as the manipulation of election district maps to either favor or disfavor a group of people (i.e. a political party, a racial/ethnic group, a socioeconomic class, etc).

1.4 Negative Racial Gerrymandering

Negative racial gerrymandering dilutes minority voting power by engineering districts in which minority ballots systematically matter less. Although minority voters retain the formal right to vote, they lack proportional influence.

Unfortunately, this practice has materialized throughout U.S. history. For instance, in response to black Americans obtaining the right to vote in 1870, Reconstruction-era Redeemer Southern Democrats wielded negative racial gerrymandering to preserve white supremacy and curtail black political power [Valelly, 2004].

This distortion produces legislatures that are less representative - often whiter and more ideologically extreme than the electorate - and weakens elected officials' incentives to respond to minority interests.

By breaking the link between voting and representation, negative racial gerrymandering undermines democratic legitimacy, erodes trust, and contributes to lower turnout and political disengagement.

1.5 The Legality of Negative Racial Gerrymandering

Negative racial gerrymandering is illegal in the United States by virtue of the Equal Protection Clause of the Fourteenth Amendment and Section 2 of the Voting Rights Act.

The Equal Protection Clause of the Fourteenth Amendment forbids denying anyone of "equal protections of the law" [United States Congress, 1868].

Section 2 of the Voting Rights Act of 1965 precludes states and political entities from wielding any "standard, practice, or procedure" to curtail the voting rights of any U.S. citizen based on race [United States Congress, 1965]. In the 1986 case of Thornburg v. Gingles, it was ruled that a North Carolina redistricting plan was in violation of Section 2 of the Voting Rights Act for engaging in negative racial gerrymandering practices - more specifically, of diluting the votes of black citizens [Supreme Court of the United States, 1986]. This landmark Supreme Court case decided that minority voters can legally challenge negative racial gerrymandering practices in redistricting plans on the grounds of violating Section 2 of the Voting Rights Act, but that the burden of proof is on them to demonstrate that their capacity to elect their preferred candidates is undermined [Supreme Court of the United States, 1986].

In the Thornburg v. Gingles case, the Supreme Court established a framework for minority voters to be able to prove negative racial gerrymandering in violation of Section 2 of the Voting Rights Act in a court of law [Supreme Court of the United States, 1986]. Under this framework, plantiffs must be able to prove the following: (1) a "geographically insular" minority group makes up the majority population of a district, (2) the minority group is "politically cohesive", and (3) the "bloc voting majority" is usually able to defeat the minority group's preferred candidates [Supreme Court of the United States, 1986]. The second and third preconditions are stated by the U.S. Supreme Court as concerning "racially polarized voting" [Supreme Court of the United States, 2023].

The U.S. Supreme Court refers to this framework as "the Gingles framework" [Supreme Court of the United States, 2023]. For the last 40 years, claims of negative racial gerrymandering in violation of Section 2 of the Voting Rights Act have been assessed via the Gingles framework [Supreme Court of the United States, 2023]. In the 2023 Allen v. Milligan decision, the U.S. Supreme Court reaffirmed the validity of the Gingles framework [Supreme Court of the United States, 2023].

1.6 Current Mid-Decade Redistricting Efforts

Starting in 2025, a coordinated wave of mid-decade redistricting efforts has emerged. The Congressional Research Service has documented that lawmakers in California, Missouri, North Carolina, Ohio, Texas, and Utah enacted ahead of the 2026 midterm elections, with additional states actively considering following in their footsteps [Whitaker and Garrett, 2026].

Texas initiated this mid-decade redistricting wave. Reacting to pressure from the Trump administration to expand the Republican congressional delegation, the Texas Legislature convened a special session and, following a House Democratic quorum break, enacted a map expected to add flip 5 Congressional seats from Democrat to Republican [Goodman, 2025]. Texas' plan faced so many accusations of negative racial gerrymandering that this escalated to the U.S. Supreme Court in LULAC v. Abbott [Supreme Court of the United States, 2025a]. It should be noted, however, that when the U.S. Supreme Court issued a stay order

for the case, the Texas map was permitted to be enacted into law [Supreme Court of the United States, 2025b].

2 Related Work

2.1 Redistricting Data Preparation

2.1.1 Constructing Racial Categories

In this paper, the Data and Democracy Lab's conventions for constructing demographic categories are adopted.

Their convention for categorizing racial groups involves adopting these group names in the ensuing order: Black, Hispanic, Asian, Other,and White [Donnay, M. and Duchin, M. and Rock, P., n.d.].

It is emphasized that anyone who identifies as black should be grouped into the black population [Donnay, M. and Duchin, M. and Rock, P., n.d.]. In other words, the black population should encompass those who are "Any-Part Black" or "Black Alone or in Combination", as opposed to just "Black Alone" or "Single-Race Black" [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Then, of the individuals left, the Hispanic population must encompass all individuals who identify as Hispanic in the same way the Black population does [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Next, of those left, the Asian population should capture all who identify as Asian or Native Hawaiian Pacific Islander, so as to represent an AAPI category [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Finally, the same logic is used to construct the AMIN, Other, and White categories [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Thereafter, the Decennial Census Redistricting PL 94-171 dataset is leveraged to generate voting age population as sums of census variables [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Subsequently, the Citizen Age Voting Population (CVAP) by Race and Ethnicity, a special tabulation from the 5-year American Community Survey, is analyzed. The problem is, CVAP data from the ACS must be analyzed at the census block level because the base unit of the Decennial Census PL 94-171 data is the census block. In the primary tables, ACS is reported at the census tract level, which presents a problem because that is a significantly higher level of aggregation than the census block [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Fortunately, the Data and Democracy Lab has introduced a solution to this problem called "discounting" [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Discounting refers to the usage of the tract-level ACS data to estimate the citizenship rates of racial groups, then applying the discount to the voting age population [Donnay, M. and Duchin, M. and Rock, P., n.d.].

An example of "discounting" provided by the Data and Democracy Lab is depicted below; in this example, block X belongs to tract Y in state Z and BCVAP is estimated by taking the citizenship rate for the group in the tract and applying it to the voting ager population in the block [Donnay, M. and Duchin, M. and Rock, P., n.d.].

$$\frac{BCVAP_{ACS}(tract\,Y)}{BVAP_{ACS}(tract\,Y)} \times BVAP_{PL}(block\,X) = BCVAP(block\,X)$$

If the denominator in an equation like the one above is too low, then the Data and Democracy Lab proposes a solution like the one below. This involves swapping in the citizenship rate given that the tract did not allow for a meaningful estimate [Donnay, M. and Duchin, M. and Rock, P., n.d.].

$$\frac{BCVAP_{ACS}(state\,Z)}{BVAP_{ACS}(state\,Z)} \times BVAP_{PL}(block\,X) = BCVAP(block\,X)$$

The Data and Democracy Lab acknowledges that this is a "crude fix", but states that they intend to release a "more sophisticated" Bayesian method to address this in the future [Donnay, M. and Duchin, M. and Rock, P., n.d.].

2.2 Ecological Inference

To reiterate what is stated in Section 1.5, in order to prove a violation of §2 of the Voting Rights Act under the Gingles framework,plaintiffs must prove that the majority votes as a bloc to defeat minority-preferred candidates [Supreme Court of the United States, 1986]. Hence, any framework performing §2 Voting Rights Act analysis must encompass identification of minority-preferred political candidates [Becker et al., 2021].

In state election return datasets, vote totals for political candidates are not computed by race, meaning that direct computation of minority-preferred candidates is impossible [Becker et al., 2021]. This is where ecological inference - the de-facto solution for determining racially polarized voting - comes in. After all, ecological inference is widely regarded as "the best and often only hope of making progress" when it comes to extracting clues about individual behavior from group or aggregate-level data [King et al., 2004].

Ecological inference (and ecological regression, its simpler predecessor) is considered a "statistical mainstay" in §2 Voting Rights Act litigation [Freedman, 1999]. U.S. Supreme Court redistricting cases in which lower-court findings relying on ecological inference were reviewed and accepted without methodological objection include Allen v. Milligan [Supreme Court of the United States, 2023], Alexander v. State Carolina State Conference of the NAACP [Supreme Court of the United States, 2024], Cooper v. Harris [Supreme Court of the United States, 2017a], Bethune-Hill v. Virginia State Board of Elections [Supreme Court of the United States, 2017b], and Abbott v. Perez [of the United States, 2018].

It should be noted, however, that in the same way all statistical inference techniques with missing data give way to uncertainty, ecological inference produces "imperfect" and "uncertain" answers [Becker et al., 2021]. Hence, it is essential to estimate the error produced by ecological inference techniques and observe how it compounds or cancels out in high-level conclusions [Becker et al., 2021].So, in addition to identifying the minority-preferred candidates, one must specify the confidence measures that the identifications are correct [Becker et al., 2021]. To check for minority control of a district and identify preferred candidates for newly proposed districts, the statewide and precinct-level vote estimates by race should be leveraged [Becker et al., 2021].

Becker et al. [Becker et al., 2021] implement a version of King's ecological inference, specifically the `ei.mD.bayes` function from eiPack which is based on the Bayesian hierarchical Dirichlet model for $R \times C$ tables proposed by King et al. in 1999 [King et al., 1999].

For each election, EI is run at the statewide level using precinct-level input tables [Becker et al., 2021]. The inputs for each precinct are the row and column sums for the $R \times C$ table of vote counts. The row sums correspond to the precinct's estimated number of adult citizens in each racial group. The column sums are the precinct's vote totals for each candidate [Becker et al., 2021].

Each ecological inference run generates a large random sample of estimated vote counts; in order to obtain the statewide estimates, these values are summed across the entire state [Becker et al., 2021]. For each racial group, the candidate with the highest average estimated vote total for an election is the preferred candidate [Becker et al., 2021]. In order to obtain a measure of confidence for the a candidate was correctly identified as the preferred candidate, repeated draws for the ecological inference distribution are obtained and the frequency by which the preferred candidate receives the most votes is recorded [Becker et al., 2021].

2.3 Ensemble-Based Outlier Analysis in Redistricting

Recent methods for analyzing and comparing district plans attempt to do so by placing a plan in the context of valid alternatives - that is, those that cut up the same jurisdiction by the same rules and with the structural features of the geography and the pattern of voting held constant [DeFord et al., 2021]. These ensembles

contain samples from the full space of plans, aiming to help compare a plan's properties to the range of possible designs [DeFord et al., 2021].

In one powerful application, ensembles have been used to conduct outlier analysis, or to argue that a proposed plan has properties that are extreme relative to the comparison statistics of alternative plans [DeFord et al., 2021]. Arguments like this have featured in a string of recent legal challenges to gerrymanders, all of which were successful at the district court or Supreme Court level [DeFord et al., 2021]. Outliers have also recieved significant attention from the U.S. Supreme Court (culminating in Rucho v. Common Cause [U.S. Supreme Court, 2019]) [DeFord et al., 2021]. Today, ensemble-based outlier analysis is considered the dominant empirical tool to evaluate gerrymandering claims in court.

The way ensemble-based outlier analysis works is that an enacted district plan π_0 is compared against a large sample of alternative plans from a neutral baseline distribution. The relative position of π_0 within the ensemble is offered as evidence of atypicality. This methodology was introduced by redistricting expert Jowei Chen in the mid-2010s [Chen and Rodden, 2013].

2.3.1 Recombination Markov Chains

Early implementations of ensemble-based outlier analysis leveraged flip-based Markov chains [DeFord et al., 2021]. However, these were discovered to be problematic for several reasons.

In 2021, Daryl DeFord, Moon Duchin, and Justin Solomon set forth recombination (ReCom), a novel family of Markov chains intended for the ensemble generation of alternative redistricting plans [DeFord et al., 2021]. Recombination is widely considered to be the standard engine for district plan ensemble generation in the redistricting space [Becker et al., 2021].

Recombination chains are considered to be superior to flip chains because they produce more compact districts, whereas flip chains are said to produce "winding, fractal-like districts" [DeFord et al., 2021]. Also, Recom performs much faster than flip becuase flip can only change the assignment label of a few nodes at a time [DeFord et al., 2021].

The way a Markov chain works is it proposes whole-district swaps via uniform spanning tree bipartitions of merged adjacent districts [DeFord et al., 2021].

2.3.2 The Mixing-Time Problem

Ensemble-based inference tests rely on the premise that the Markov chain has run long enough for the sampled plans to be approximately representative of the target distribution. The validity of any p-value derived from a single long chain depends on this premise holding. Existing work defends the premise to convergence diagnostics - running multiple chains from different starting points, comparing summary statistics across chains, and monitoring autocorrelation in test statistics over chain length.These diagnostics are necessary but not sufficient because they cannot certify convergence and they offer no formal bound on the bias they fail to detect.

The underlying difficulty is that the mixing time of ReCom on real-world precinct graphs remains unknown, and it is not known to be polynomial in any rigorous sense. Cannon et al. establish that although the spanning tree distribution targeted by ReCom is well-defined and the kernel approximately converges to it, there exists a residual gap between the measures of the uniform spanning tree approximation actually implemented and the exact spanning tree [Cannon et al., 2022a].They also state that polynomial mixing remains conjectural even on well-behaved graphs [Cannon et al., 2022a]. Likewise, Daryl DeFord, Moon Duchin, and Justin Solomon characterize convergence on practical graphs as an empirical question rather than a settled theoretical result [DeFord et al., 2021].

Hence, every ensemble-based p-value reported in redistricting literature is only valid if the chain has mixed. In Rucho v. Common Cause, ensemble-based outlier analysis was called into question since the underlying chain's convergence properties are not formally established [U.S. Supreme Court, 2019].

2.3.3 Approaches to the Mixing Time Problem

Three responses to the mixing problem has appeared in literature.

The most dominant approach in applied work is more compute - that is, longer chains, more starting points, and more elaborate diagnostics. However, this is not a true solution because the result is still not valid unless the chain is mixed and no amount of empirical convergence diagnostic can rule out a slow-mixing pathology that the diagnostic is not designed to detect.

Cannon et al. introduced the second approach: short bursts [Cannon et al., 2022a].Short bursts sacrifice ensemble representativeness to optimize over plans satisfying particular criteria [Cannon et al., 2022a].Although short bursts are excellent for detecting extreme plans, they are not intended for inferential testing of an enacted plan against a neutral baseline [Cannon et al., 2022a].

The third approach to address this problem is to leverage an inferential framework that does not require mixing at all. In 1989, Julian Besag and Peter Clifford introduced two such frameworks for generalized Monte Carlo significance tests [Besag and Clifford, 1989]. The first is the serial method, where the test statistic's null distribution is conducted from a single chain via time-reversal arguments [Besag and Clifford, 1989]. The second method and the one adopted in this paper is the parallel method, where independent chains launched from a common state produce an exchangeable ensemble whose ranks are exactly uniform on the null [Besag and Clifford, 1989]. Neither of the two methods require the chain(s) to have mixed; the only requirement is reversibility of the kernel with respect to the target measure [Besag and Clifford, 1989]. Julian Besag and Peter Clifford's methods are well-established in spatial statistics, ecology, and image analysis, but have not been applied to redistricting ensembles.

2.3.4 Quantifying Minority Opportunity

The most influential framework for quantifying minority was introduced by Becker et al. in 2021 [Becker et al., 2021]. This approach combines ecological inference to estimate each racial group's candidate preferences, recency-weighted aggregation cycles across elections to smooth idiosyncratic single-election noise, and confidence weighted aggregation that downweights elections with diffused rather than polarized preferences [Becker et al., 2021]. The output is a district-level "effectiveness score" $s_d^g \in [0, 1]$ which measures the weighted share of recent elections in which a racial group g's preferred candidate won [**?**].

This "effectiveness score" is then thresholded to count "opportunity districts" [Becker et al., 2021]. For example, a district is considered to be "Latino effective" if S_D^L exceeds a cutoff (typically 0.5 or 0.6), and the count of the districts is the plan's opportunity score [Becker et al., 2021]. This thresholding step is the pressure point of existing literature. There are two major issues with this. The first is that the threshold is theoretically arbitrary; in other words, there is no principled reason to choose 0.5 over 0.6 or vice versa. The second issue is that the threshold is fragile near the cutoff point. For example, a district with $s_g^d = 0.59$ counts for nothing while a district with $s_g^d = 0.61$ counts in full, even though these two districts are almost identical electorally. This fragility is especially evident in situations when the effectiveness scores cluster near the cutoff.

The natural alternative introduced in each paper is to stop thresholding altogether and allowing the per-district win probability p_d^g enter the plan-level function directly: $O_g(\pi) = \sum_d p_d^g(\pi)$, the expected number of districts where racial group g's preferred candidate won. This functional has been informally suggested in the literature, but it has not been adopted as the basis for a formal inferential test.

3 Contribution

The components surveyed in Section 2 (that is, the redistricting data preparation, $R \times C$ ecological inference, and ensemble generation) exist as separately developed and well-documented tools. However, there is currently no published framework that integrates the three into one pipeline where posterior uncertainty from

ecological inference propagates through to plan-level opportunity scoring and to the inferential test against an enacted plan. This paper supplies that integration. There are four specific contributions this paper offer.

3.1 A Unified, Modular Pipeline

An end-to-end pipeline is introduced that takes raw publicly available data sources (Decennial Census PL 94-171 Redistricting Data, ACS CVAP tabulations, state election returns, TIGER/LINE geographies) and produces a calibrated probability of minority electoral opportunity for every district in every plan. Modular in this context refers to the fact that each layer can be swapped out independently. For instance, improvements in the ecological inference layer do not require redoing the data engineering and ensemble generation & inference layer.

3.2 Uncertainty Propagation Rather Than Point Estimates

The ecological inference stage exports per-precinct, per-group, per-candidate octile quantiles of the posterior count distribution, and the ensemble stage Monte Carlo samples within those octile bins when scoring each candidate plan.The plan-level opportunity score therefore not only reflects variability among plans but also variability of voter-behavior in each plan. In contrast, in common practice, point estimates from ecological inference are passed downstream as if they were measured quantities.

3.3 A Probabilistic Notion of Minoirty Opportunity

The opportunity functional $O_g = \sum_d p_d^g(\pi)$ satisfies anonymity, district-additivity, monotonicity, and normalization.

3.4 Exact Inference Under Unknown Mixing Time

The mixing time problem associated with Recombination is sidestepped altogether via the Besag-Clifford parallel construction, which yields exact p-values regardless of how thoroughly the chain has mixed. Hence, $§§$2 Voting Rights Act analysis will no longer depend on unverifiable convergence claims and does this without modifying the recombination kernel itself.

4 The Framework

4.1 Data Engineering Layer

4.1.1 Data Sources

The data engineering layer ingests the following raw inputs:

1. State District Plan Shapefile
2. Decennial Census PL 94-171 Redistricting Dataset
3. Citizen Voting Age Population by Race and Ethnicity in a Special Tabulation from the ACS 5-Year Estimates 4. State Census Blocks Shapefile (TIGER/LINE)
4. State Voting Tabulation District Shapefile
5. State VTDs Election Data

4.1.2 Data Engineering Layer Overview

Figure 1 depicts the pipeline through which these raw data inputs are combined. The ingestion stage standardizes formats and projections. Construction of demographic categories follows the methodology outlined in Section 2.1.1. A geographic crosswalk allocates blocks to precincts by spatial intersection, after which keys are harmonized across the demographic and electoral sources, the unified dataset is assembled, and a final schema enforces derived proportion columns and consistent column names. Validation checks confirm that population sums match between sources and that no precinct is left without a district assignment.

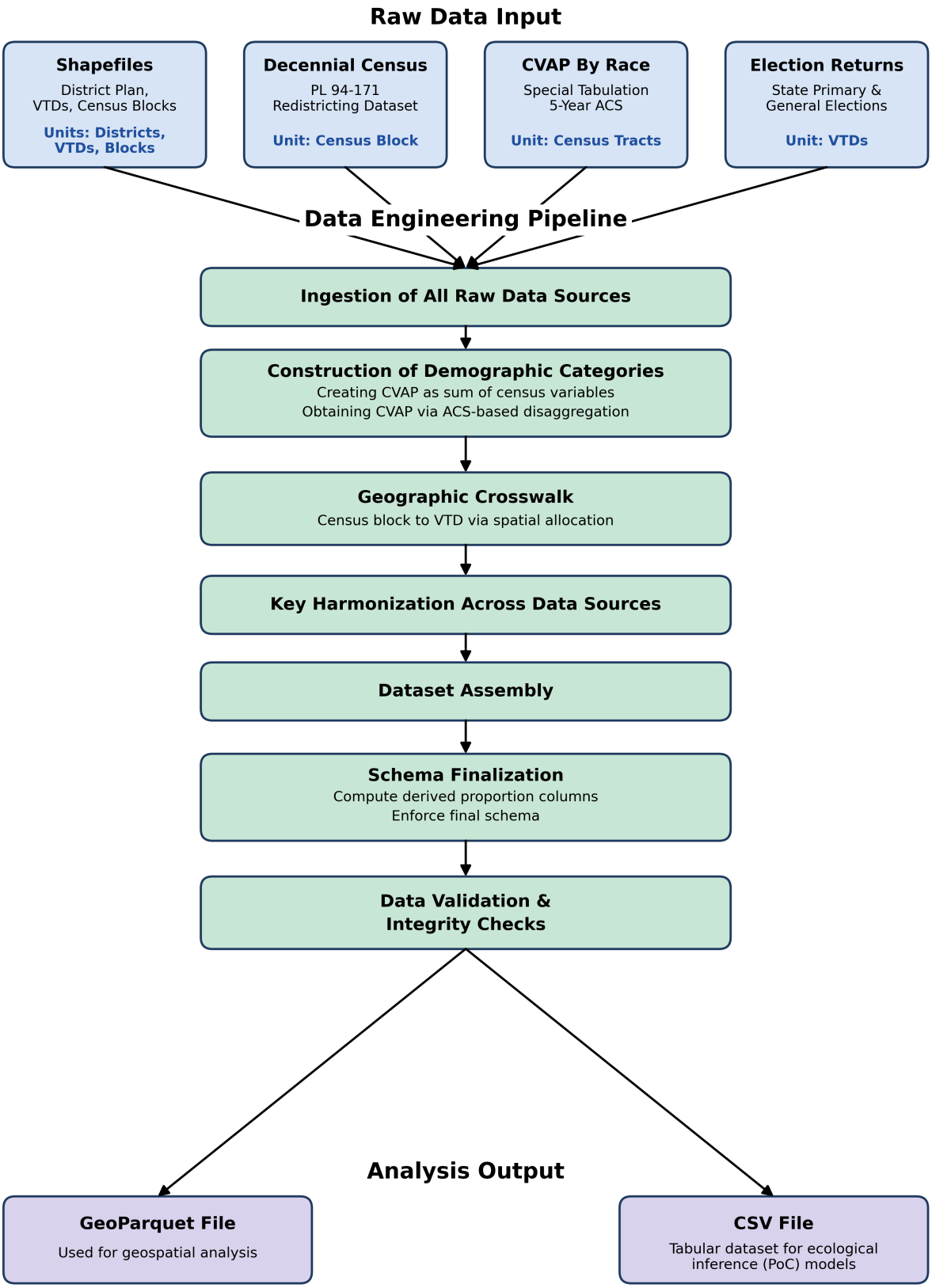


Figure 1: Overview of the data engineering layer.

4.2 Ecological Inference Layer

The ecological inference layer's first inferential stage converts observed precinct-level election returns and citizen voting-age population (CVAP) composition into group-conditional posterior distributions over vote choice. These distributions are the substrate on which all downstream Voting Rights Act opportunity scoring is built: they identify the minority-preferred candidates in each contest, they provide precinct-level posterior summaries of group-specific vote counts (used to construct district-level support estimates over the ensemble of plans), and they supply the raw scores to which a per-mode, per-group logistic calibration is fit.

Inputs and the Inferential Target

The dataset generated in the data engineering layer is a precinct-level table covering all of Texas. For each precinct p there is a vector of CVAP shares $X_p = (X_{p,1}, \ldots, X_{p,R}) \in \Delta^{R-1}$ across $R = 5$ racial groups (Hispanic, Black, White, Asian, and American Indian/Native) together with per-candidate vote counts and total turnout for each contest in our election inventory. Mixed primaries are split by party so each fitted election has a single, party-specific non-voting category; the Republican "Uncommitted" line in the 2024 Presidential primary is dropped because it admits no candidate-level racial identification.

Because the secret ballot makes the joint distribution of race and vote choice unobservable, we recover it by ecological inference: a model that reconciles the two observed marginals — racial composition by precinct and vote totals by precinct — to produce a credible posterior over within-precinct, group-specific support.

Model and Inference

We use the row-by-column (RxC) hierarchical Multinomial-Dirichlet Bayesian model of **?**, the standard specification for redistricting EI since **?**. Let C index the columns: one per candidate plus an *Abstain* column, so column totals sum to CVAP rather than to ballots cast. For each precinct p and each group r, define a latent simplex $\beta_{p,r,\cdot} \in \Delta^{C-1}$ giving group r's probabilities over the C outcomes in precinct p. The hierarchical specification is

$$\beta_{p,r,\cdot} \mid \alpha_r \;\sim\; \text{Dirichlet}(\alpha_r), \qquad \alpha_r \;\sim\; \pi(\alpha_r), \qquad r = 1, \ldots, R,\;\; p = 1, \ldots, P,$$

where $\pi(\alpha_r)$ is the weakly-informative hyperprior implemented by the `RowByColumnEI` class of `pyei` [**?**]. Conditional on $\{\beta_{p,r,c}\}$ and the (fixed) group fractions X_p, the precinct-level outcome probabilities are the group-share-weighted mixture

$$\theta_{p,c} \;=\; \sum_{r=1}^{R} X_{p,r}\, \beta_{p,r,c}, \qquad c = 1, \ldots, C,$$

and the observed column counts $V_{p,\cdot}$ — votes by candidate plus the abstention residual — follow

$$V_{p,\cdot} \mid \theta_p \;\sim\; \text{Multinomial}(N_p, \theta_p), \qquad N_p \;=\; \text{CVAP}_p.$$

Treating CVAP as the multinomial denominator (rather than ballots cast) is deliberate: it ties the abstention rate to the same demographic composition used to identify candidate support, so heterogeneity in turnout across groups is absorbed into the posterior over β rather than being silently projected out by re-normalising the columns. For each contest we fit (4.1)–(4.3) independently; precincts with $N_p = 0$ or with no votes recorded for the contest are excluded, and contests retaining fewer than 50 precincts after filtering are dropped.

Posterior inference uses the No-U-Turn Sampler [**?**] with 1{,}000 retained draws per chain after 100{,}000 tuning steps, a single chain, and a fixed random seed; sampling is performed in single precision to keep the posterior tensor (draws $\times$ precincts $\times$ groups $\times$ candidates) within memory. Per-contest traces are checkpointed to NetCDF so that the postprocessing stage can be rerun without re-sampling.

Posterior Summaries

Three quantities are exported per contest. Let S index the retained MCMC draws and let $\beta^{(s)}_{p,r,c}$ denote the s-th draw.

Statewide preferences. Group r's statewide support for candidate c in draw s is the CVAP-weighted average of within-precinct probabilities,

$$\hat{S}^{(s)}_{r,c} \;=\; \frac{\sum_p X_{p,r}\, N_p\, \beta^{(s)}_{p,r,c}}{\sum_p X_{p,r}\, N_p}.$$

The candidate of choice is $c^\star_r = \arg\max_c \overline{S}_{r,c}$ where $\overline{S}_{r,c} = S^{-1}\sum_s \hat{S}^{(s)}_{r,c}$. We report $\overline{S}_{r,c^\star_r}$ alongside the fraction of draws under which $c^\star_r$ is the modal preference,

$$\pi_r \;=\; \frac{1}{S}\sum_{s=1}^{S} \mathbf{1}\Big[\arg\max_c \hat{S}^{(s)}_{r,c} \;=\; c^\star_r\Big].$$

The latter quantity, written `prob` in the downstream tables, supplies the posterior credibility that group r's preferred candidate is in fact $c^\star_r$ and is used to weight contests in the cohesion calculation.

Precinct mean counts. For each (precinct, group, candidate) we record the posterior mean estimated count

$$\overline{n}_{p,r,c} \;=\; X_{p,r}\, N_p\, \frac{1}{S}\sum_{s=1}^{S} \beta^{(s)}_{p,r,c}.$$

Precinct count octiles. The method Becker et al. leverage to record precinct-level probabilistic information in a condensed form is adopted here [Becker et al., 2021].This method allows distributions to be recorded at every step of the Markov chain and involves compressing a histogram into octiles, storing only eight "bars" as opposed to "dozens of hundreds" [Becker et al., 2021].

Within-precinct posterior uncertainty is preserved for downstream Monte Carlo by exporting eight octile quantiles of the group-specific count posterior, evaluated at the probabilities $\mathcal{Q} = \{k/8 : k = 1, \ldots, 8\}$:

$$Q_{p,r,c}(q) \;=\; \mathrm{Quantile}_q\Big(\big\{X_{p,r}\, N_p\, \beta^{(s)}_{p,r,c}\big\}_{s=1}^{S}\Big), \qquad q \in \mathcal{Q}.$$

The ensemble layer treats each successive pair of octiles as a uniform bin and samples linearly within it, recovering an empirical eight-bin approximation to the full posterior distribution of $X_{p,r}N_p\beta_{p,r,c}$. The implicit floor $q = 0$ is set to zero, an approximation that biases the simulated draw floor downward only when the posterior places non-negligible mass near zero counts.

These three artefacts are pivoted from the long, draw-aware format produced by the sampler into the wide, composite-key format `{CVAP\_KEY}.{election}\_{candidate}.{permille}` consumed by the ensemble layer, where `CVAP_KEY` maps the EI group label to the corresponding CVAP column (Hispanic $\rightarrow$ HCVAP, Black $\rightarrow$ BCVAP) and `permille` is the integer $\lfloor 1000 \cdot q \rfloor$. We additionally emit a single-row CSV of in-group cohesion weights $w \in \{1.0, 0.75, 0.5\}$ that the ensemble layer applies to elections in which the candidate of choice is, respectively, a co-ethnic of the group, a co-ethnic of one of the two minority groups but not both, or neither.

Calibration of VRA-Opportunity Scores

The ensemble-layer scoring function (Section 5) returns, for each district d, each model mode $m \in \{\text{statewide}, \text{equal}, \text{district}\}$, and each subgroup $g \in \{\text{Black}, \text{Latino}, \text{Neither}\}$, a raw VRA-opportunity score $\rho_{d,m,g} \in [0,1]$ constructed as a weighted share of minority-preferred-candidate votes. Raw scores are internally consistent across plans but are not directly interpretable as probabilities of the corresponding opportunity outcome; their ordinal information must be mapped to a calibrated probability scale before being aggregated across modes and subgroups via the Venn-overlap formula used in Section 5.

We calibrate $\rho_{d,m,g}$ by fitting one univariate logistic regression per (m,g) cell — nine models in total. The training data are produced by scoring the enacted Texas congressional plan with the ensemble-layer machinery in `return_raw` mode, harvesting $\{\rho_{d,m,g}\}_{d=1}^{38}$, and pairing each district with a binary label $y_{d,m,g}$ derived from a benchmark general-election outcome:

$$y_{d,m,\text{Black}} \;=\; \mathbf{1}[\text{black-preferred candidate wins district } d]\,,$$

with $y_{d,m,\text{Latino}}$ defined analogously and $y_{d,m,\text{Neither}} = 1 - \max(y_{d,m,\text{Black}}, y_{d,m,\text{Latino}})$. Multiple benchmarks may be pooled; raw scores depend only on the enacted plan and so are computed once and reused across benchmarks. For each (m,g) we then fit

$$\Pr(y_{d,m,g} = 1 \mid \rho_{d,m,g}) \;=\; \sigma(\alpha_{m,g} + \beta_{m,g}\,\rho_{d,m,g})\,, \qquad \sigma(z) = \frac{1}{1+e^{-z}},$$

via Newton-Raphson on the penalised log-likelihood

$$\ell(\alpha,\beta) \;=\; \sum_d \Big[y_d \log\sigma(\alpha+\beta\rho_d) + (1-y_d)\log\big(1-\sigma(\alpha+\beta\rho_d)\big)\Big] - \tfrac{\lambda}{2}(\alpha^2+\beta^2),$$

with $\lambda = 10^{-6}$ for numerical stability. When a calibration cell contains only one observed class (a common occurrence for the Neither subgroup under tightly clustered benchmarks) we revert to the identity calibration $\alpha_{m,g} = 0$, $\beta_{m,g} = 1$, propagating the raw score unchanged rather than allowing complete separation to drive $|\beta|$ to infinity.

The fitted parameters $\{(\alpha_{m,g}, \beta_{m,g})\}$ are written to a nine-row table that the ensemble layer consults whenever it requires a calibrated opportunity probability. Recalibration is inexpensive, requires no re-sampling of the EI posterior, and is the natural point at which to ingest new benchmark data when additional general elections become available.

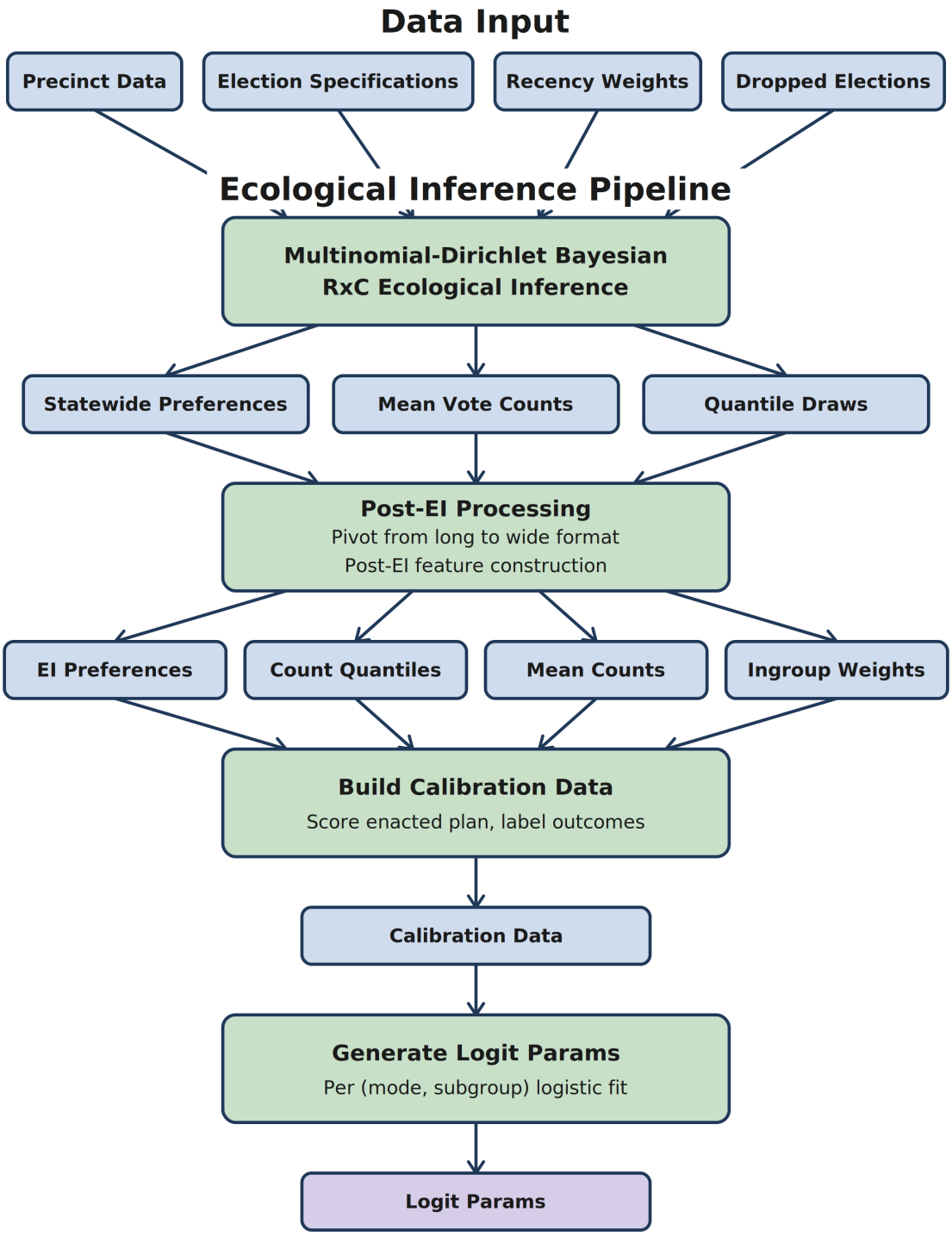


Figure 2: Overview of the ecological inference layer.

4.3 Ensemble Generation & Inference Layer

4.3.1 Setup and Notation

The notation of Daryl DeFord, Moon Duchin, and Justin Solomon's foundational Recombination Markov chain paper is adopted here. A graph $G = (V, E)$ is constructed where V is a set of vertices for each geographic unit (precincts, in this case) and E is a set of edges between those geographic units that are geographically adjacent [DeFord et al., 2021]. Hence, a redistricting plan is a partition of the nodes of V into subsets that induce connected components of G [DeFord et al., 2021]. In this manner, redistricting can be reframed as a graph partitioning problem [DeFord et al., 2021]. Equivalently, a redistricting plan is an assignment of each node to one of k districts via a labeling map $V \to \{1, \cdots, k\}$ [DeFord et al., 2021].

So, to restate the above, $V = \{v_1, \ldots, v_K\}$ denotes the precincts in the state, with rook-adjacency graph $G = (V, E)$. Let π be a redistricting plan be a partition $V \to 1, \cdots, K$ of K districts. Let the enacted redistricting plan being analyzed be denoted as π_0.

In dual graphs derived from real-world data, nodes are weighted with populations or demographic data, which is represented with functions $w : V \to R$ [DeFord et al., 2021]. This notation enables the efficient expression of constraints [DeFord et al., 2021]. Let ϵ be a small population deviation greater than zero [Cannon et al., 2022b].

A condition that bounds population deviation can be modeled as

$$(1-\epsilon)\frac{\sum_V w(v)}{k} \leq |V_i| \leq (1+\epsilon)\frac{\sum_V w(v)}{k}$$

[DeFord et al., 2021].

Let $\Pi_k(\epsilon)$ denote the set of redistricting plan where every district induces a connected subgraph and the population balance condition holds for every district d.

Let ρ denote the spanning-tree measure on $\Pi_k(\varepsilon)$,

$$\rho(\pi) \;\propto\; \prod_{d=1}^{K} \tau\big(G[\pi^{-1}(d)]\big),$$

where $\tau(H)$ is the number of spanning trees of subgraph H [Cannon et al., 2025].

The hypothesis we test is

$$H_0 : \pi_0 \sim \rho_{\mathcal{C}},$$

where $\rho_{\mathcal{C}}$ is ρ restricted to a constraint subset $\mathcal{C} \subseteq \Pi_k(\epsilon)$ encoding compactness and county-respect requirements [DeFord et al., 2021]. The test statistics are scalar functionals $T : \Pi_k(\varepsilon) \to \mathbb{R}$ measuring minority opportunity.

Despite strong empirical performance, the mixing time of ReCom Markov chains on realistic precinct graphs is not theoretically characterized [Cannon et al., 2022a]. As with many high-dimensional combinatorial Markov Chain Monte Carlo methods, polynomial mixing is conjectured but not proven [Levin et al., 2009], and convergence rates may be slow in practice. We sidestep this concern by adopting the inference procedure described below; upon leveraging this procedure, validity holds exactly regardless of mixing.

4.3.2 The Minority Opportunity Functional

For each protected minority group g, each plan π, and each district d, let $p_d^g(\pi) \in [0, 1]$ denote the model-based probability that group g's preferred candidate wins in district d.

The probabilities $\{p_d^g(\pi)\}$ are produced by the scoring layer within the ecological inference layer, which combines a $R \times C$ ecological-inference model of group-specific candidate preferences with recency-weighted

election histories under three weighting schemes (statewide, equal, and district). The opportunity functional is defined as the sum of district-level win probabilities, as demonstrated below.

$$O_g(\pi) \;=\; \sum_{d=1}^{K} p_d^g(\pi),$$

and a coalition functional crediting each district to whichever group has the higher win probability,

$$O_{\text{joint}}(\pi) \;=\; \sum_{d=1}^{K} \max\{p_d^B(\pi),\, p_d^L(\pi)\}.$$

$O_g(\pi)$ admits an axiomatic justification (anonymity, district-additivity, monotonicity, normalization) given in §2. As an estimand it is the expected number of districts in which group g's preferred candidate prevails, given the win probabilities supplied by the scoring layer.

To express deviation from the *De Grandy* proportional-representation baseline [*Johnson v. De Grandy*, 512 U.S. 997 (1994)], let $\bar{m}_g$ denote the statewide CVAP share of group g, and define the diversity-calibrated functional

$$F_g(\pi) \;=\; O_g(\pi) - K\,\bar{m}_g,$$

with F_{joint} defined analogously using $\bar{m}_B + \bar{m}_L$. F_g is a translation of O_g that centers the demographic baseline at zero; $F_g(\pi) < 0$ indicates expected opportunity below proportionality. We adopt F_g as the primary test statistic, and report O_g and the ensemble-standardized

$$Z_g(\pi) \;=\; \big(O_g(\pi) - \mathbb{E}_{\rho_C}[O_g]\big)\big/\sqrt{\mathrm{Var}_{\rho_C}(O_g)}$$

as auxiliary quantities, with $\mathbb{E}_{\rho_C}$ and Var_{ρ_C} estimated from the spoke ensemble of §4.4.

4.3.3 Axiomatic Justification of the Minority Opportunity Functional

Here, it is argued that the minority opportunity functional O_g is the unique plan functional that satisfies four natural properties for a measure of expected group electoral opportunity.

Axiom 1 (Anonymity): For any permutation σ of district labels $\{1, \ldots, K\}$, $T(\pi_\sigma) = T(\pi)$, where π_σ is the plan obtained by relabeling districts.'

Axiom 2 (District Additivity): $T(\pi) = \sum_{d=1}^{K} t(d, \pi)$, where $t(d, \pi)$ is a *district contribution* depending only on the precincts and covariates of district d under π.

Axiom 3 (Monotonicity): If π and π' are plans with the same district contribution function and $p_d^g(\pi) \leq p_d^g(\pi')$ for every d, then $T(\pi) \leq T(\pi')$.

Axiom 4 (Normalization): $0 \leq T(\pi) \leq K$, with the bounds attained when $p_d^g(\pi) = 0$ for all d and $p_d^g(\pi) = 1$ for all d, respectively.

Proof. Anonymity follows because the sum is invariant under permutation of summands. District-additivity holds with $t(d, \pi) = p_d^g(\pi)$. Monotonicity holds term-by-term. For normalization, $p_d^g(\pi) \in [0, 1]$ implies $0 \leq \sum_d p_d^g(\pi) \leq K$, with the bounds attained at the stated configurations.

For the converse, A2 forces $T(\pi) = \sum_d t(d, \pi)$. The hypothesis that $t(d, \pi)$ depends on π only through $p_d^g(\pi)$ writes $t(d, \pi) = \phi(p_d^g(\pi))$ for some ϕ. Monotonicity (A3) requires ϕ weakly increasing; normalization (A4) at the boundary cases requires $\phi(0) = 0$ and $\phi(1) = 1$. ■

The proposition narrows the choice of opportunity functional to monotone reweightings of win probabilities. We adopt the identity reweighting $\phi(p) = p$, yielding O_g, because (i) it has the cleanest interpretation as expected number of opportunity districts and (ii) any non-trivial ϕ introduces additional modeling choices that obscure the connection to the calibrated probabilities produced in §@ref(sec:calibration).

The Neutral Ensemble: ReCom with VRA-Relevant Constraints

Proposals are generated by ReCom [DeFord, Duchin, and Solomon, 2021]. At each step the kernel (a) selects an adjacent district pair uniformly at random, (b) draws a uniform spanning tree on the merged subgraph, and (c) selects an edge whose removal yields a balanced bipartition. Combined with always-accept and the *allow-pair-reselection* variant of `bipartition_tree`—which re-rolls the district pair when no balanced cut exists rather than rejecting the proposal outright—the kernel P approximately satisfies detailed balance with respect to ρ on the constrained state space [DeFord, Duchin, and Solomon, 2021; Cannon et al., 2022]. We rely below on this approximate reversibility; see §4.6 for a discussion of the residual exactness gap.

The constraint set $\mathcal{C}$ comprises the population-balance requirement embedded in $\Pi_K(\varepsilon)$ together with two state-relative thresholds:

(i) *Compactness.* Total cut-edges, $\mathrm{CE}(\pi) = |\{e \in E : \pi(u) \neq \pi(w) \text{ for } e = \{u, w\}\}|$, must satisfy

$$\mathrm{CE}(\pi) \;\leq\; s_c \cdot \mathrm{CE}(\pi_0), \qquad s_c = 1.5.$$

The slack s_c is conventional in the redistricting literature [Mattingly et al., e.g., Herschlag et al., 2020]. Setting $s_c = \infty$ recovers ReCom's implicit spanning-tree compactness preference alone.

(ii) *County splits.* Let $\mathrm{CS}(\pi)$ denote the number of counties whose precincts span two or more districts. We impose

$$\mathrm{CS}(\pi) \;\leq\; s_g \cdot \mathrm{CS}(\pi_0), \qquad s_g = 1.2.$$

This constraint addresses the standard expert-witness objection that ensemble plans achieve elevated minority-opportunity scores by carving counties more aggressively than feasible legislative drafts. The 20% slack is chosen to preserve the constraint's bite while admitting sufficient mobility for the chain to mix; tighter slacks substantially slow mixing on the Texas precinct graph.

Both thresholds are scaled to π_0 rather than imposed in absolute terms, rendering $\mathcal{C}$ state-relative: the null distribution we compare π_0 against consists of plans that are no less compact and no more county-splitting than π_0 itself, scaled by the slacks.

We do *not* impose a hard contiguity validator. ReCom's spanning-tree mechanism guarantees that every accepted proposal is contiguous by construction; imposing contiguity additionally as a validator would cause the chain to reject π_0 as initial state, because the within-district subgraphs of certain enacted Texas CDs are not graph-connected under the precinct-vintage rook adjacency. This is an artifact of urban CD geometry interacting with rook adjacency rather than a substantive discontiguity.

Besag-Clifford Exact Inference

Conventional ReCom analyses report tail-rank statistics from a single long chain and rely on convergence diagnostics to defend their validity. We instead apply the *parallel method* of Besag and Clifford [1989, Proposition 3.3], which yields exact p-values under H_0 regardless of the kernel's mixing time.

Construction. Starting from π_0, run P forward for L_{hub} steps to obtain a *hub* state

$$X^* \;\sim\; P^{L_{\mathrm{hub}}}(\pi_0, \cdot).$$

From X^*, generate M conditionally independent *spokes*

$$Y^{(1)}, \ldots, Y^{(M)} \;\overset{\text{i.i.d.}}{\sim}\; P^{L_{\mathrm{spoke}}}(X^*, \cdot).$$

Under H_0 and the reversibility of P with respect to $\rho_{\mathcal{C}}$, the joint law of (π_0, X^*) is identical to that of (X^*, π_0), so π_0 given X^* has the same distribution as each $Y^{(j)}$ given X^*. Consequently $(\pi_0, Y^{(1)}, \ldots, Y^{(M)})$ is exchangeable conditional on X^*, and the rank of any test statistic T evaluated at π_0 within this $(M+1)$-tuple is uniform on $\{1, \ldots, M+1\}$.

The lower-tail Besag-Clifford p-value is therefore

$$\widehat{p}_{\text{low}}(T) \;=\; \frac{\left|\{j : T(Y^{(j)}) \le T(\pi_0)\}\right| + 1}{M+1}, \tag{5}$$

and satisfies $\Pr_{H_0}\big(\widehat{p}_{\text{low}}(T) \le \alpha\big) \le \alpha$ for every $\alpha \in \{1/(M{+}1), 2/(M{+}1), \ldots\}$. The upper-tail expression is analogous. Crucially, the argument requires only reversibility of P with respect to $\rho_{\mathcal{C}}$; it places no requirement on L_{hub} or L_{spoke}.

Hyperparameter selection. The validity of (5) is independent of L_{hub}, L_{spoke}, and M, but each affects either resolution or power.

Resolution. The minimum attainable lower-tail p-value under (5) is $1/(M+1)$. We adopt $M = 1000$, giving a floor of $\approx 10^{-3}$ — the modal choice in published redistricting work [Mattingly group; MGGG case studies]. Raising M to 10,000 (floor $\approx 10^{-4}$) is appropriate for litigation/expert-witness contexts and is supported by the implementation without modification beyond compute.

Power. L_{hub} and L_{spoke} govern dependence structure within the spoke ensemble. Small L_{hub} leaves X^* and hence the spokes correlated with π_0, depressing power against alternatives near π_0 in the state graph; small L_{spoke} leaves spokes correlated with one another, inflating estimator variance for $\mathbb{E}_{\rho_{\mathcal{C}}}[O_g]$ and $\mathrm{Var}_{\rho_{\mathcal{C}}}(O_g)$. We take $L_{\text{hub}} = 2000$ and $L_{\text{spoke}} = 500$. The asymmetric ratio reflects that X^* is generated once and shared across all spokes (so the unit cost of L_{hub} is amortized) while the per-spoke cost scales linearly in L_{spoke}.

5 Case Study: Texas Congressional Districts

5.1 Data Engineering Layer Implementation

1. Texas PLANC2333 District Plan Shapefile
2. Decennial Census PL 94-171 Redistricting Dataset
3. Citizen Voting Age Population by Race and Ethnicity in a Special Tabulation from the ACS 5-Year Estimates
4. Texas Census Blocks Shapefile (TIGER/LINE)
5. Texas Voting Tabulation District Shapefile
6. State VTDs Election Data

Table 1 below summarizes the resulting statewide demographic shares.

Table 1: Texas demographics by racial group across population, voting-age population, and citizen voting-age population.

Racial Group	Total Population	VAP	CVAP
Hispanic	34.7%	35.6%	32.6%
Black	13.5%	13.0%	12.9%
White	46.2%	43.2%	48.9%
Asian	5.0%	6.1%	4.8%
American Indian	0.7%	1.3%	0.7%

As shown in Table 2 below, the election inventory comprises six 2024 statewide contests.

Table 2: Texas statewide elections used in the case study.

Office	Stage	Year
President	General	2024
President	Primary	2024
U.S. Senate	General	2024
U.S. Senate	Primary	2024
Railroad Commissioner	General	2024
Railroad Commissioner	Primary	2024

5.2 Ecological Inference Layer Implementation

We fit the $R \times C$ model of §@ref(sec:ei-layer) to each contest in Table @ref(tab:tx-elections) and report posterior support estimates, preference identifications, and per-precinct count posteriors at the octile resolution.

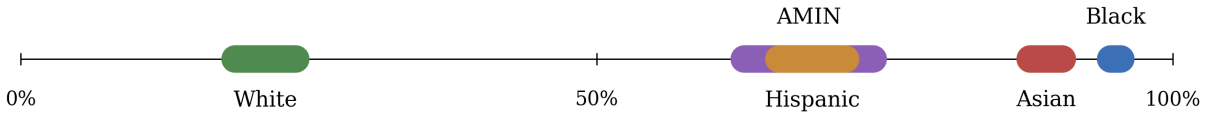


Figure 3: Highest and lowest EI point estimates for each racial group's support of the 3 Democratic nominees in statewide general elections: White (19–24%), AMIN (63–74%), Hispanic (66–72%), Asian (88–90%), and Black (95–96%).

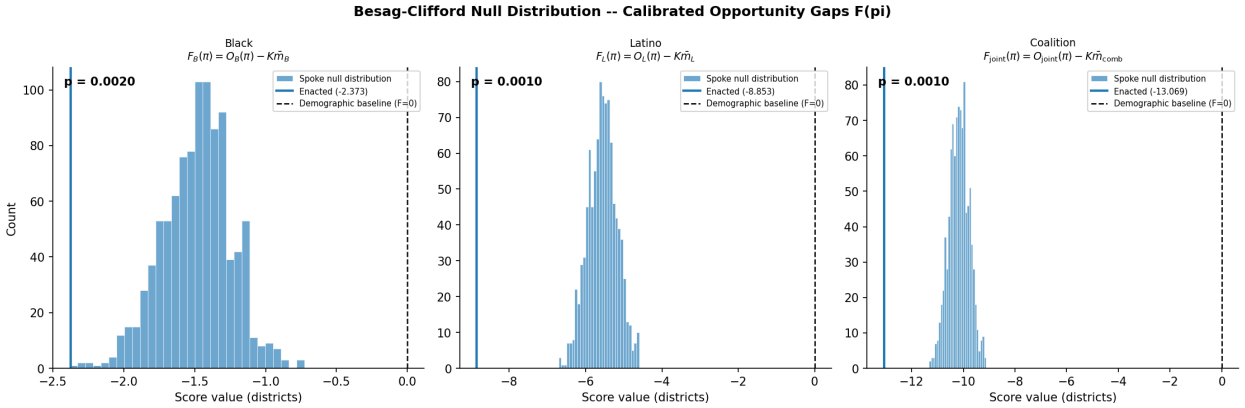


Figure 4: f distributions

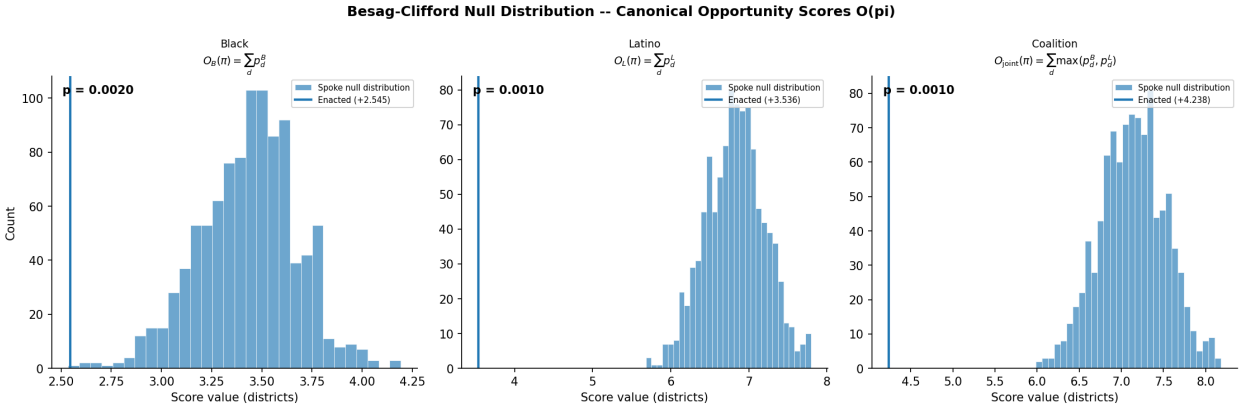


Figure 5: O dist

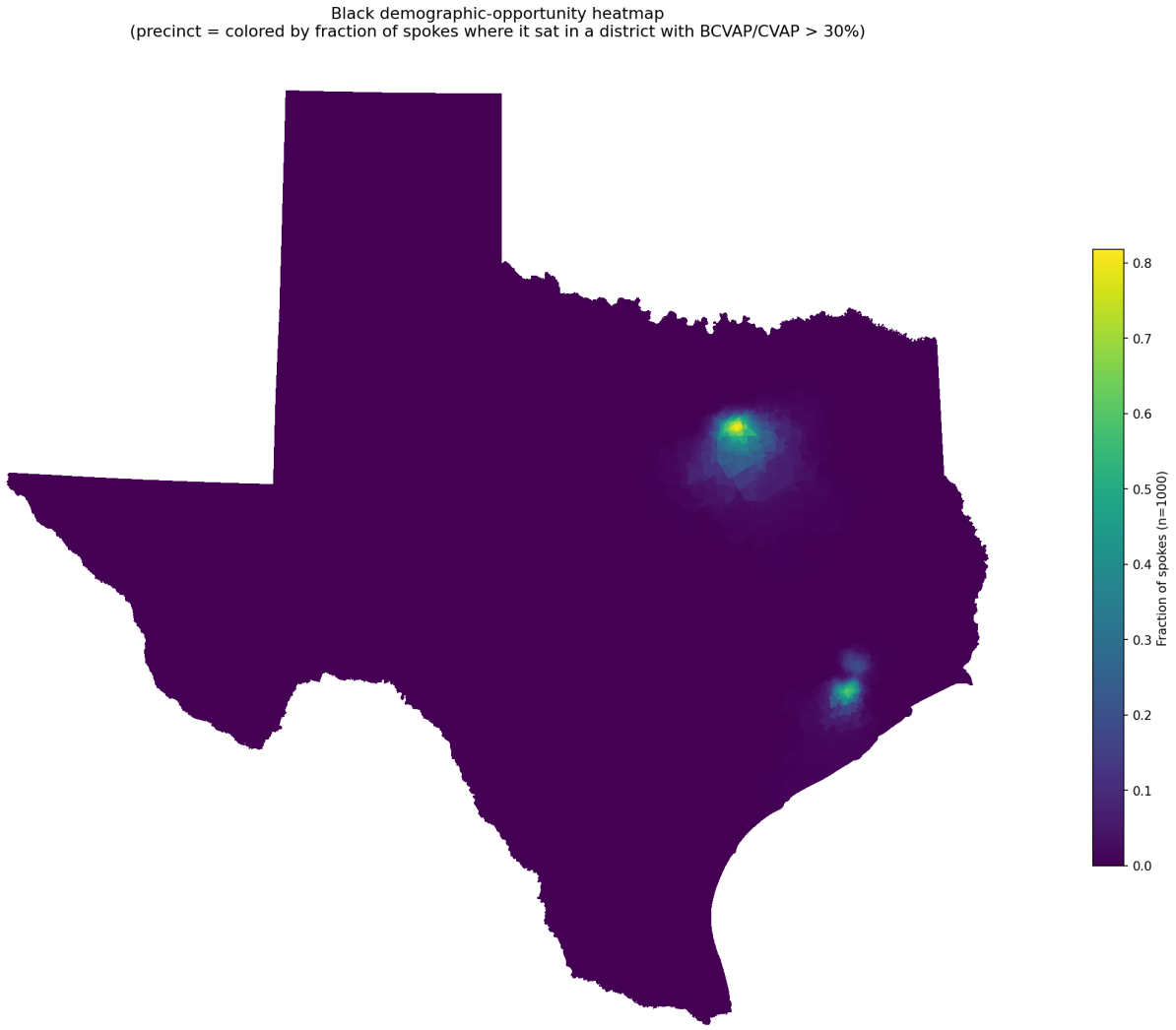
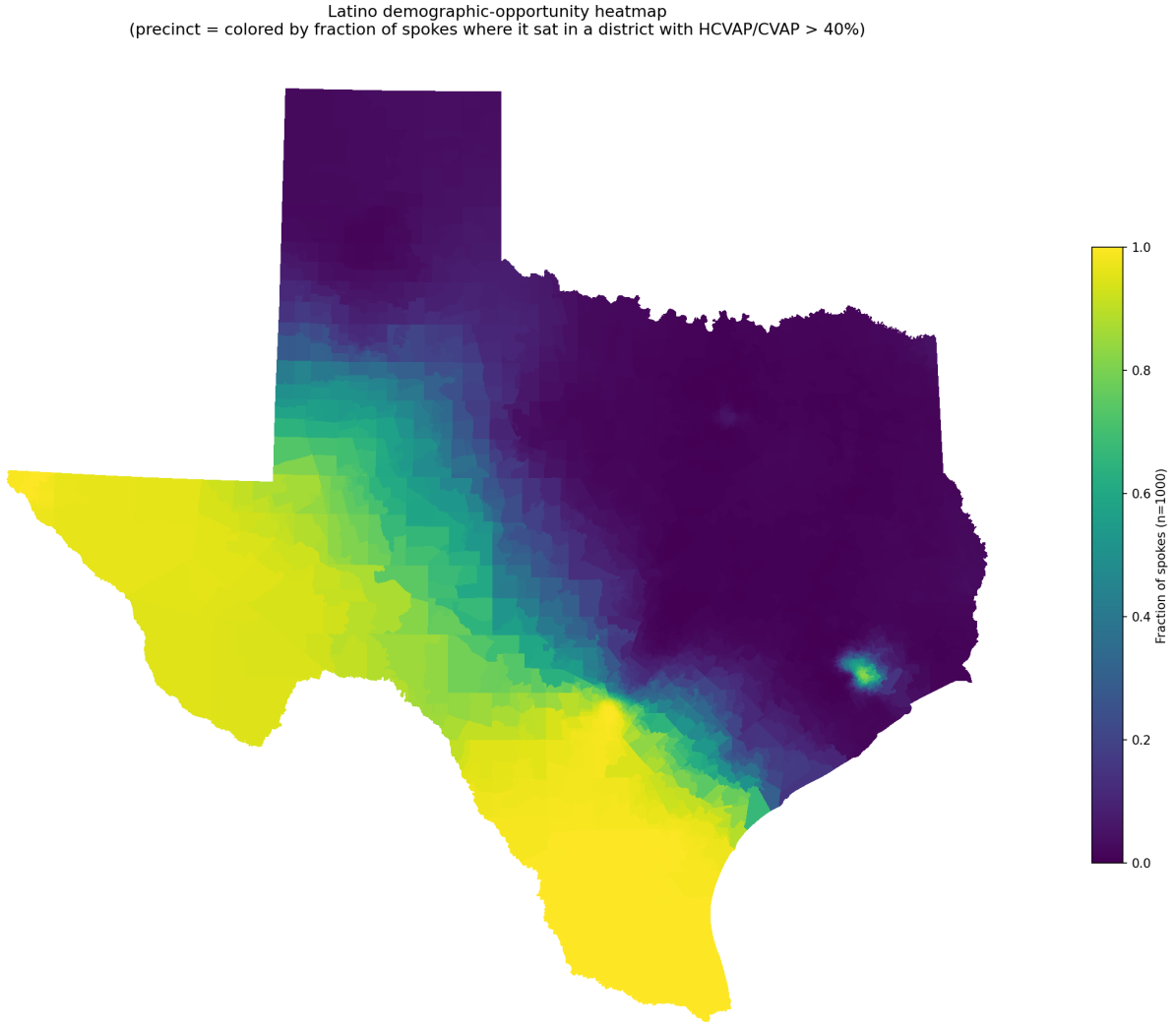
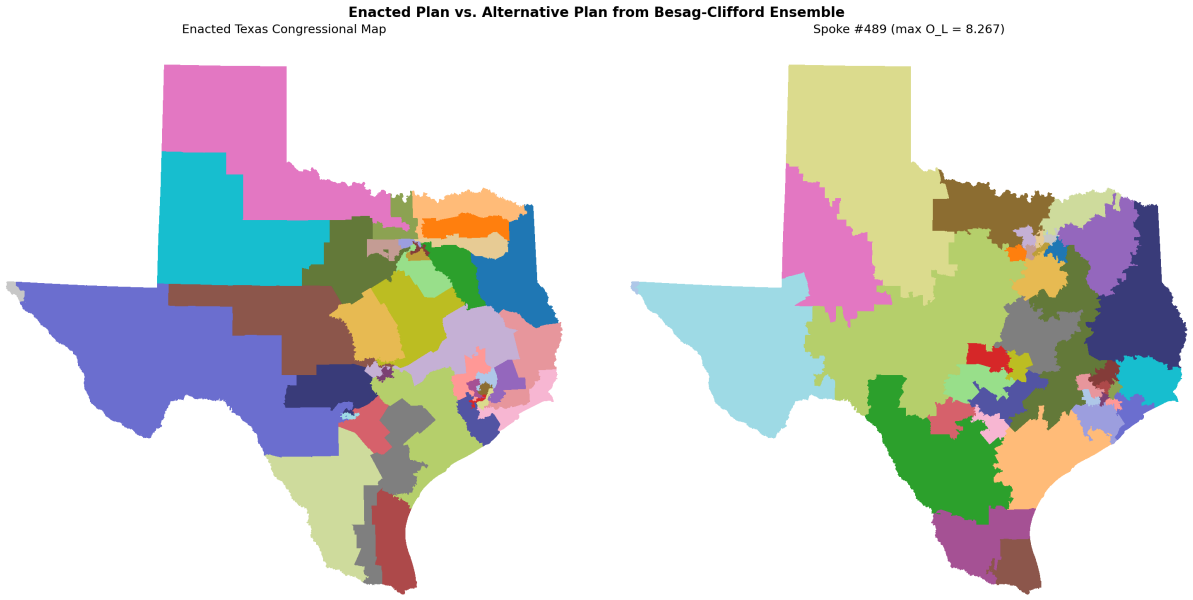


Figure 6: black opportunity heatmap


Latino demographic-opportunity heatmap
(precinct = colored by fraction of spokes where it sat in a district with HCVAP/CVAP > 40%)
1.0
0.8
0.6
0.4
0.2
0.0
Fraction of spokes (n=1000)


Enacted Plan vs. Alternative Plan from Besag-Clifford Ensemble
Enacted Texas Congressional Map
Spoke #489 (max O_L = 8.267)

6 Discussion

6.1 Limitations

This framework relies on several assumptions including approximate reversibility of the ReCom kernel, correct specification of the ecological inference model, accuracy of CVAP estimates, and representativeness of the selected elections. These limitations should be considered when interpreting results.

7 Conclusion

This paper introduces an exact inferential framework for redistricting analysis that eliminates reliance on Markov chain mixing assumptions. By combining Besag-Clifford inference with uncertainty-aware ecological modeling, a principled and statistically valid approach to evaluating redistricting plans is set forth.

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